

Module 7

Advanced Git for Collaboration

Learning objectives

- Create and switch feature branches
- Merge branches and resolve conflicts
- Understand submodules for shared IP
- Use git blame and bisect conceptually

Prerequisites

Module 6 (add, commit, push)

Learning path

Learning Path

Diagram: Learning Path
(render with mmdc when available)
flowchart LR
A[Branches] --> B[Merge]
B --> C[Conflicts]
C --> D[Submodules]
D --> E[blame & bisect]
E --> F[Module 8]

Course context

Team labs use branches per feature

Conflicts happen when two people edit same lines

Verification IP may live in submodules

Core topics

Self-check

```
$ ./scripts/module7.sh --check
```

Terminal: module_check

```
█[0;36m=====█[0m
█[0;36mModule 7: Environment self-check█[0m
█[0;36m=====█[0m

█[0;32m[OK]█[0m git available: git version 2.43.0
█[0;32m[OK]█[0m module7 directory exists: /mnt/d/proj/universal-verification-methodology/learn_unix_
█[0;32m[OK]█[0m conflicts/sample_conflict.txt exists

█[0;32m[OK]█[0m All required checks passed.
```

Feature branch

```
git checkout -b feature/lab2  
git merge main
```


Merge conflict format

```
$ head -15 module7/examples/conflicts/sample_conflict.txt
```

Terminal: conflict_sample

This file shows how Git marks merge conflicts (for reference only).
In a real conflict, Git inserts these markers into your file.

<<<<<< HEAD

This is the version from the current branch (e.g. main).
Keep this if you want the main-branch change.

=====

This is the version from the branch being merged (e.g. feature/x).
Keep this if you want the feature-branch change.

>>>>>> feature/x

To resolve: choose one version (or combine both), then delete the markers and the line you don't want

Branches

```
$ git help branch 2>/dev/null | head -8
```

Terminal: git_branch_help

GIT-BRANCH(1)

Git Manual

GIT-BRANCH(1)

NAME

git-branch - List, create, or delete branches

SYNOPSIS

git branch [--color[=<when>] | --no-color] [--show-current]
[-v [--abbrev=<n> | --no-abbrev]]

Practice

Exercises

Create branch in git_demo

Resolve sample_conflict.txt on paper

Try git submodule update --init

Summary & next steps

Branching and merging

Conflicts and submodules

Next: Module 8 — end-to-end workflow

Next: Module 8: End-to-End Project Workflow